

Predicting Urban Micromobility Trip Distance Using Linear and Neural Network Models

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Abstract—This study models urban micromobility trip distance using a large real-world dataset from Austin, Texas, comparing linear regression, single-layer perceptrons, and multi-layer perceptrons (MLPs). The response is log-transformed, and predictors are standardized for stable optimization. Results show that linear models achieve limited accuracy, while a shallow ReLU-activated MLP significantly improves performance. A final GPU-based mini-batch gradient descent (MBGD) model further enhances predictive accuracy, demonstrating the value of stochastic training for nonlinear mobility modeling.

Index Terms—Micromobility, neural networks, regression, deep learning, transportation analytics

I. INTRODUCTION

Shared micromobility systems, including bicycles, e-bikes, and electric scooters, have become integral components of urban transportation networks. These services provide flexible, short-distance mobility and often complement existing public transit by facilitating efficient first- and last-mile travel. As adoption continues to grow, cities increasingly rely on data-driven approaches to understand usage patterns and support planning, deployment, and regulatory decisions.

Trip distance is an especially important behavioral metric, reflecting how riders use micromobility infrastructure across temporal and operational contexts. Its determinants—such as trip duration, time of day, vehicle type, and local geography—can interact nonlinearly, which limits the expressiveness of classical linear models. This motivates the use of more flexible learning approaches capable of capturing richer structure in the data.

The objective of this study is to predict the log-transformed distance of micromobility trips using operational and temporal features from a large real-world dataset. The modeling framework compares multiple linear regression, a single-layer perceptron (SLP), and several multi-layer perceptron (MLP) architectures. A final GPU-accelerated mini-batch training regime is also evaluated to examine how modern optimization strategies affect model performance and generalization.

A. Related Work

Machine learning has shifted from symbolic reasoning to predictive modeling driven by large datasets and computational advances [1]. Deep learning expanded this progress by enabling multi-layer architectures capable of representing highly nonlinear patterns [2]. Foundational work on backpropagation

[3] and universal approximation [4] provides theoretical justification for using shallow MLPs in regression tasks.

Neural networks have been widely applied to transportation research, including traffic-flow prediction [5] and city-scale ridership modeling [6]. These studies highlight the value of nonlinear models for capturing behavioral patterns in human mobility, motivating the present work.

B. Data Description

The dataset originates from Austin's Open Data Portal [7] and spans over 15 million micromobility trips from April 2018 to April 2022. Each record provides start and end timestamps, reported distance, duration, vehicle type, and council district identifiers.

To avoid structural shifts caused by later operational changes or the COVID-19 pandemic, we restrict analysis to weekday trips during 2018, yielding roughly 1.67 million observations.

Numerical fields containing non-numeric characters were parsed using regular expressions. Observations with non-positive distance or duration were removed, as were redundant identifiers. Missing categorical entries were imputed via mode. Timestamps were mapped to Unix time, and the response was log-transformed to mitigate right skew.

The log-transformed origin–destination matrix (Figure 1) shows strong spatial variation, with District 9 having the highest trip volume and districts 1, 3, and 5 forming secondary mid-volume connections.

Ten predictors describing temporal, operational, and contextual attributes were used (Table I).

TABLE I
VARIABLES USED IN MODELING

Type	Name	Description
Response	Log Trip Distance	Natural log of distance (miles)
Predictor	Trip Duration	Seconds
Predictor	Month	1–12
Predictor	Hour	0–23
Predictor	Day of Week	0=Mon,...,6=Sun
Predictor	Start District	Council district
Predictor	End District	Council district
Predictor	Year	2018 indicator
Predictor	veh_scooter	Scooter indicator
Predictor	veh_bicycle	Bicycle indicator
Predictor	start_ts	Unix start time
Predictor	end_ts	Unix end time

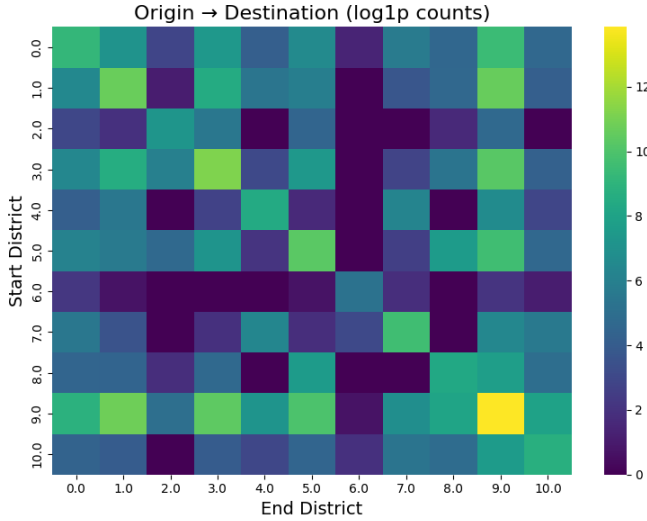


Fig. 1. Origin → Destination heatmap (log1p counts).

II. METHODS

The modeling task is formulated as the estimation of the conditional mean of the log-transformed trip distance. For each trip i , let $x_i \in \mathbb{R}^p$ denote the vector of predictors and let y_i denote the corresponding log-distance. All models considered here seek to learn a parametric function $f(x_i; \theta)$ that minimizes the empirical risk

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i; \theta))^2,$$

where θ denotes all learnable parameters. Model performance is assessed using a held-out validation set during training and a separate test set for final evaluation.

A. Data Splitting and Standardization

After preprocessing, the dataset was partitioned into training (70%), validation (15%), and test (15%) sets. To prevent information leakage, standardization parameters—means and standard deviations of the predictors—were computed strictly from the training data and subsequently applied to the validation and test sets. This ensures that all downstream model estimates and predictions rely only on information available at training time.

B. Multiple Linear Regression

Multiple linear regression (MLR) serves as a transparent baseline model. The standardized predictors are arranged into a design matrix X , and the model assumes a linear relationship

$$\hat{y} = X\beta,$$

where β is estimated via ordinary least squares. Because standardization places all predictors on a common scale, coefficient magnitudes provide direct insight into feature importance. However, this model cannot capture interactions or nonlinear effects.

C. Single-Layer Perceptron

The single-layer perceptron (SLP) retains the linear structure of MLR but replaces closed-form estimation with gradient-based optimization. The model computes

$$\hat{y} = XW^\top + b,$$

where W and b are optimized using the Adam algorithm. This formulation allows the SLP to share optimization dynamics with deeper networks, isolating the effect of model capacity from the effect of gradient-based training. Early stopping on the validation loss prevents unnecessary computation and reduces the risk of overfitting.

D. Multi-Layer Perceptron

The multi-layer perceptron (MLP) extends the SLP by introducing one or more hidden layers and nonlinear activation functions. A one-hidden-layer MLP computes

$$h = \phi(XW_1^\top + b_1), \quad \hat{y} = hW_2^\top + b_2,$$

where ϕ is a nonlinear activation function such as ReLU. Deeper MLPs stack multiple layers of this form, allowing the network to represent complex, high-order interactions. During training, the validation loss is monitored at each epoch, and training halts when improvement falls below a predefined tolerance for several consecutive epochs.

The MLP experiments systematically vary activation function choice, network depth, learning rate, and dropout probability. This enables a structured evaluation of how architectural and optimization choices affect convergence, generalization, and computational efficiency.

E. Mini-Batch Gradient Descent and GPU-Accelerated Training

Because full-batch gradient descent produces only one update per epoch, it can converge slowly and may become trapped in sharp minima. To address this, the final model was additionally trained using mini-batch gradient descent (MBGD). At each iteration t , the gradient is approximated by a batch B_t :

$$\nabla_{\theta} \mathcal{L}_{B_t}(\theta) = \frac{1}{|B_t|} \sum_{i \in B_t} \nabla_{\theta} \ell(f(x_i; \theta), y_i).$$

This introduces controlled stochasticity, leading to more frequent parameter updates and potentially better generalization.

Training was implemented using PyTorch's `DataLoader` with batch size 8192 and performed on an NVIDIA CUDA GPU when available. All tensors were moved to the GPU, and forward- and backward-pass computations were executed on-device. Random seeds were fixed for Python, NumPy, and PyTorch to ensure reproducibility. Early stopping used patience 5 and a tolerance `min_delta` = 10^{-3} to accommodate the noisier validation trajectory typical of MBGD.

III. RESULTS

All performance metrics reported in this section refer to mean squared error (MSE) computed using the log-transformed response variable. Thus, all references to “MSE” should be interpreted as “MSE on the log scale.” Results are organized into baseline comparisons, activation-function experiments, depth and learning-rate sweeps, dropout ablation, and mini-batch GPU-accelerated training.

A. Baseline Comparisons

Both multiple linear regression (MLR) and the single-layer perceptron (SLP) achieve test-set MSE values near 1.566, confirming that optimization alone cannot compensate for the limited representational capacity of linear models. By contrast, the best-performing full-batch MLP—a one-hidden-layer network with width 64, ReLU activation, and learning rate 0.01—achieves a substantially lower MSE of 0.8475.

TABLE II
BASELINE MODEL COMPARISON (TEST MSE, LOG SCALE)

Model	Test MSE
Multiple Linear Regression	1.5658
Single-Layer Perceptron	1.5658
Best MLP (1 layer, ReLU, lr=0.01)	0.8475

SLP training and validation curves (Figure 2) reveal slow but steady improvement without overfitting.

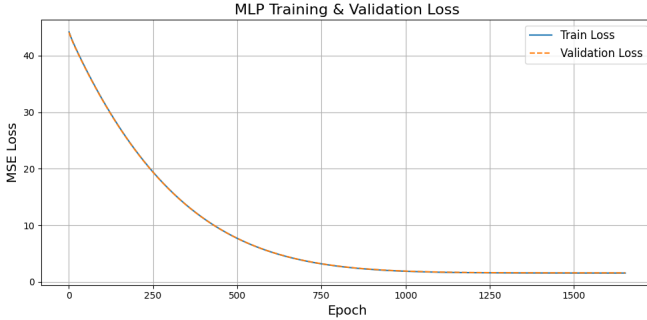


Fig. 2. Training and validation loss for the SLP (full-batch training).

B. Activation Function Experiments

Four activations—ReLU, leaky ReLU, ELU, and Softplus—were evaluated under identical architectural and optimization settings. ReLU produced the lowest test MSE, while ELU and Softplus exhibited early stagnation and substantially poorer performance.

TABLE III
ACTIVATION FUNCTION COMPARISON (1 LAYER, WIDTH 64, LR=0.01)

Activation	Test MSE
ReLU	0.8475
Leaky ReLU	0.8505
ELU	2.6795
Softplus	2.1798

Figure 3 shows the learning curves for the best ReLU-based MLP.

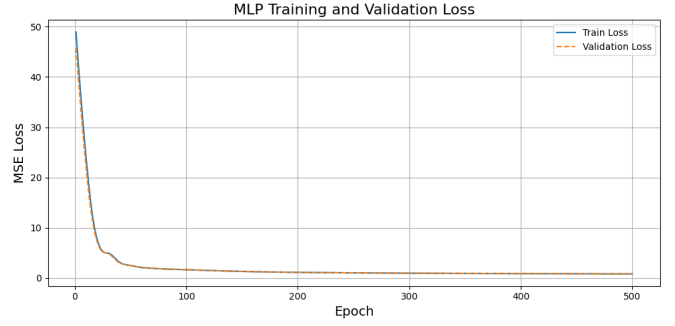


Fig. 3. Training and validation loss for the best-performing MLP (full-batch training).

C. Depth and Learning Rate Experiments

A grid search over depths $\{1, 2, 3\}$ and learning rates $\{0.01, 0.003, 0.001\}$ revealed strong interactions between architecture size and learning rate. While depth-1 networks showed stable training at all learning rates, deeper models became unstable at $lr = 0.01$ and required smaller step sizes. CPU training time increased with depth, and no deeper architecture outperformed the single-layer MLP.

TABLE IV
DEPTH $\times$ LEARNING RATE RESULTS WITH CPU TRAINING TIME

Depth	LR	Val MSE	Test MSE	Epochs	CPU Time (sec)
1	0.01	0.8414	0.8475	500	82.7
1	0.003	1.2166	1.2260	500	90.6
1	0.001	2.0828	2.0705	500	91.8
2	0.01	4.2337	3.4945	25	4.9
2	0.003	0.8673	0.8753	500	100.1
2	0.001	1.5919	1.5848	500	119.9
3	0.01	4.2763	3.7016	26	6.6
3	0.003	0.9204	0.9286	500	122.6
3	0.001	1.0944	1.1062	500	121.8

D. Dropout Experiments

Dropout probabilities in $\{0.0, 0.1, 0.2, 0.3, 0.5\}$ were evaluated. Although dropout reduced training time by triggering earlier stopping, it consistently degraded performance relative to the no-dropout baseline.

TABLE V
DROPOUT COMPARISON (1 LAYER, WIDTH 64, ReLU, LR=0.01)

Dropout	Val MSE	Test MSE	Epochs	CPU Time (sec)
0.0	0.8414	0.8475	500	93.9
0.1	1.8371	1.8085	89	25.7
0.2	1.8915	1.8263	89	28.6
0.3	1.8945	1.8321	90	22.6
0.5	1.7379	1.7347	109	30.2

E. Mini-Batch Gradient Descent (MBGD) and GPU Training

The final model was trained using MBGD (batch size 8192) with Adam ($\text{lr} = 0.01$) on a CUDA GPU and early stopping (patience 5, $\text{min_delta} = 10^{-3}$). MBGD yielded faster early-epoch improvement and achieved a significantly lower test MSE of 0.7178 compared to full-batch training (0.8475).

TABLE VI
MBGD + GPU TRAINING RESULTS (BATCH SIZE = 8192)

Metric	Value
Best Validation MSE	0.6886
Test MSE	0.7178
Early-Stopping Epoch	44
Training Time (sec)	478.31

TABLE VII
MBGD TRAINING AND VALIDATION LOSS (10-EPOCH INTERVALS)

Epoch	Train Loss	Val Loss
1	6.5562	1.7899
10	0.8682	0.8631
20	0.7398	0.7370
30	0.7228	0.7423
40	0.6912	0.7145
44 [†]	0.6884	0.6886

[†]Early stopping triggered after five consecutive epochs without validation improvement $\geq 10^{-3}$.

Figure 4 visualizes this behavior, with initial noise in the validation trajectory giving way to gradual improvement that culminates in the lowest observed validation loss at epoch 44.

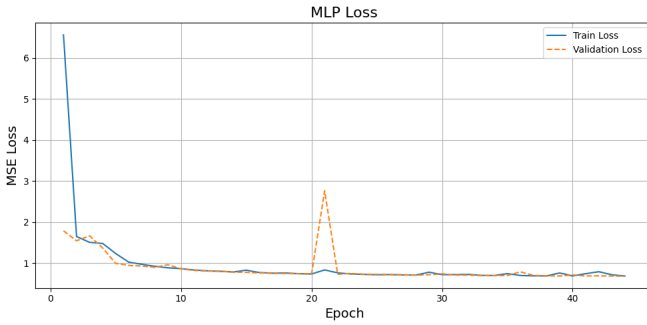


Fig. 4. Training and validation loss for the MBGD + GPU-trained MLP.

F. Overall Summary of Experimental Results

The linear models performed poorly and were unable to capture the patterns in the data, while models with nonlinear structure did noticeably better. Among the activation functions tested, ReLU was by far the most effective. Adding more hidden layers did not help—deeper networks ran longer but did not produce more accurate predictions. Dropout, which is often useful for regularization, consistently reduced performance in this setting. The strongest results came from training the MLP using mini-batch gradient descent on a GPU, which produced the lowest test error and improved noticeably over the full-batch version of the same model.

IV. CONCLUSION

This study compared linear regression, single-layer perceptrons, and multi-layer neural networks for predicting micromobility trip distance in Austin. The results show that a straightforward one-hidden-layer MLP, trained with mini-batch updates, offers the best balance of accuracy and efficiency for predicting micromobility trip distances in this dataset.

Preprocessing choices such as log transformation and standardization proved essential for stable training. Hyperparameter experiments highlighted the importance of activation function and learning rate, whereas additional depth and dropout provided limited benefit. Although predictive accuracy was the main focus, interpretability remains relevant; SHAP (SHapley Additive exPlanations) offers a model-agnostic, statistically grounded approach for attributing predictions to individual features and represents a promising extension of this work.

Future research could incorporate more detailed spatial features, explore temporal modeling, or scale to deeper architectures using high-performance computing resources. These extensions would further support data-driven approaches to urban mobility planning and policy.

APPENDIX

All project code and notebooks are available at:
<https://github.com/bryant-willoughby/ST-507-Data-Science-Analytics-Using-Python>.

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